

Correction of smooth scan distortions using strain gradients

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1 Motivation

Non-linear sample motion and small distortions of the microscope scan present a formidable challenge to ptychographic reconstructions. The difficulty stems from strong dependence of diffraction patterns and the reconstructed object on probe positions. For a given probe and object, small variations in the scan can make a big difference. "Small" here carries some nuance with it. On one hand it can mean a small perturbation of a particular probe position, or a small local distortion of the scan about a probe position. The former is well suited for corrections by simply using gradients of probe positions. But the latter is more difficult, since small distortions across the scan can accumulate, making the required displacements for corrections very large. Gradient descent based reconstructions try and match diffraction patterns by making small changes. A key requirement is the existence of a "reasonable" (perhaps somewhat smooth and without false minima) path from the initial state to the optimal state. For large corrections this assumption may not hold true¹ when simply using perturbation of individual probe positions in each iteration.

For a more robust method of updating positions, understanding the different components of scan distortions is necessary. Here, two assumptions are used:

1. Scan corrections in the linear regime are used exclusively.
2. The initial object learning rate is assumed to be high enough so that the object is "tethered" to the scan initially.

Next, noting that at each diffraction pattern the probe position only sees the object it encounters, the scan correction must be extracted from the object itself. Distortion of the scan then corresponds to local distortions of the object, which is assumed to be strongly coupled to the scan. All linear transforms of the object can be represented as a combination of strains and translations:

$$O_{lin}(x) = O(Mx + C) \quad (1)$$

where M is a matrix that consists of strains and rotation, and C is a translation.

The translation of the object can be replaced by a translation of the probe position, and is the naive scan gradient currently used. If all neighboring probe positions are close to correct, and consequently the object is largely undistorted, just moving the given probe position is necessary. Hence perturbations of probe positions by small amounts about their correct positions, which does not distort the object significantly, should be easily corrected this way. Manual affine transforms may help to provide such a starting point.

Nonlinear global distortions, on the other hand, require accounting for errors in neighboring probe positions, which are assumed to induce local distortions of the object. The matrix M accounts for such distortions, and the strain gradient presented below is a method proposed to address this.

2 Strain gradient

Infinitesimal linear transforms of the object, given a fixed center (probe position) can be modeled as:

$$O_{sg}(\varepsilon) = O - \sum_{i,j} \varepsilon_{ij}(x_j - x_{j0}) \frac{\partial O}{\partial x_i}, \quad \varepsilon_{ij} = 0 \quad (2)$$

¹Try changing the size of the scan and see for yourself how drastically diffraction patterns change!

where x_{j0} denotes the specific probe position being considered. This computation must be done separately at each probe position, and is natural extension to the way the forward model is usually computed.

In order to obtain strain gradients, simply evaluate the forward model using O_{sg} instead of O , with all $\varepsilon_{ij} = 0$. This results in an unchanged forward model, but enables computation of gradients with respect to ε_{ij} using `autodiff`. Gradient computation yields four additional components, similar to the scan in size. These quantify variations from distorting the object and, as a consequence, the diffraction patterns at each probe position. In essence, perturbations that stretch diffraction patterns are evaluated.

3 Iteration update

Scan strain gradients are only half the story. The corresponding iteration update step is the other half that is more nuanced, and a simple and somewhat optimal² approach is discussed here. Once the strain gradients are scaled to an appropriate update step size (not discussed here), the strain update must be converted to a displacement update that is applied to both object and scan:

1. Interpolate the strains smoothly over the object domain to obtain a grid the size of the object.
2. Use Fourier transforms of the strains to obtain displacements that produce the computed strains, i.e. invert the strains in the Fourier domain.
3. Interpolate and update both the object and scan with the resulting displacements.

3.1 Fourier inversion of strains

Inverting strains may be done in several ways, here an easy to implement and robust method based on Fourier transforms is used. First, we define differential operators (operating by convolution) on a discrete grid:

$$D_i(x) = \begin{cases} -0.5, & x_i = -1 \text{ and } x_{j \neq i} = 0 \\ 0.5, & x_i = 1 \text{ and } x_{j \neq i} = 0 \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

And, the Laplacian, in Fourier space,

$$L_{\mathcal{F}} = \sum_i \mathcal{F}[D_i]^2 \quad (4)$$

where $\mathcal{F}[D_i]$ are the Fourier transforms of the real space discrete derivative operators. Note that it is possible that $L_{\mathcal{F}} = 0$ at some points.

Next, given strain updates $\Delta\varepsilon_{ij}$ interpolated to a grid, the displacements can be found as:

$$u_i = \mathcal{F}^{-1} \left[\frac{1}{L_{\mathcal{F}}} \sum_j \mathcal{F}[D_j] \mathcal{F}[\Delta\varepsilon_{ij}] \right] \quad (5)$$

where, the inversion is done via first finding the Laplacian of each component of the displacements (by computing the gradient in Fourier space), and then inverting the Laplacian. For points where $L_{\mathcal{F}} = 0$, their inverses can be replaced by zeros. This is simply because the discrete Laplacian in this form cannot sample any object with those frequencies, and which are high enough to be ignored, since they are unrealistic in reconstructions. Furthermore, due to numerical approximations and inaccuracies, small imaginary components present in the displacements need to be removed.

Ideally, if the components of displacement field are differentiable:

$$\frac{\partial \Delta\varepsilon_{xx}}{\partial y} = \frac{\partial \Delta\varepsilon_{xy}}{\partial x} \text{ and } \frac{\partial \Delta\varepsilon_{yx}}{\partial y} = \frac{\partial \Delta\varepsilon_{yy}}{\partial x} \quad (6)$$

but these may not hold in real reconstructions, and hence the the Fourier inversion method may not necessarily be optimal or perfectly accurate. Iterative fitting or other forms of estimation may yield better results.

²Optimal here refers to a part of the update where least squares optimal estimation of displacements are computed given local strains. Note that this claim is based on a back of the envelope calculation done long back, and needs verification!

3.2 Initial and final interpolations

Since the initial strain gradients are defined at probe positions, which may have an arbitrary geometry, appropriate interpolation is needed to estimate strain updates for the grid used in the Fourier inversion for object and scan updates. Scattered point interpolation available in **MATLAB**, **numpy** or other similar languages/packages can be used for this. For the strains, extrapolation outside the convex hull can be simply set to zero, since the displacements will remain continuous. For displacements on the other hand, to maintain continuity, nearest neighbor or linear interpolation is more suitable.

Once the displacements u_i are found, both the object and scan must be updated. To update the probe positions, simply add the interpolated displacement at the old probe position:

$$x_{new} = x_{old} + u_{interp}(x_{old}) \quad (7)$$

And, similarly, to update the object:

$$O_{new}(x + u) = O_{old}(x) \quad (8)$$

In order to compute $O_{new}(x)$, interpolate $O_{old}(x)$ but rather than use x as the points for interpolation, replace them with $x + u$. This may require the use of scattered point interpolation since $x + u$ will not lie on a regular grid. The resulting interpolant takes the value $O_{old}(x)$ at the point $x + u$. Finally, to obtain the distorted object needed, evaluate the interpolant at x so that O_{new} is sampled on the original grid.

4 Conclusion

Hence we conclude the procedure to obtain the updated scan and object. Ideally, it may be beneficial to distort the probe as well, but given that the aperture is fixed, such an update must be constrained to the phase and is difficult to implement in a straightforward and stable manner. With small enough scan updates, and large enough probe updates, such complications are not necessary.

References